LAKNIDU THILAKAWARDHANA

DEVOPS & SOFTWARE ENGINEER

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PROFILE SUMMARY

DevOps and Software Engineer with proven expertise in orchestrating scalable cloud infrastructure, automated CI/CD pipelines, and high-performance full-stack applications. Experienced in Docker, Kubernetes, AWS, and modern microservices using .NET, NestJS, and Next.js. Dedicated to system reliability, infrastructure-as-code, and crafting frictionless developer workflows.

WORK EXPERIENCE

Intern Software and Devops Engineer - Koombiyo delivery **Jan 2024 - Present**

- Architected and managed scalable cloud deployments on AWS with automated CI/CD pipelines using GitHub Actions, cutting release cycles by 55%.
- Implemented micro-frontend architectures with Next.js and optimized backend rendering pipelines, resulting in a 40% reduction in latency.
- Containerized core services with Docker and orchestrated workloads on Kubernetes, maintaining 99.98% system uptime.

EDUCATION

BSc (Hons) in Computer Science (Second Upper Class Honours) **2022 - 2025**

NSBM

PROJECTS

Work Projects

Smart Home IoT Management System

C++, Next.Js, ExpressJS, Node, PostgreSQL, React Native

Smart Home IoT Management System | Full-Stack & IoT Development

Designed and developed a smart home IoT management system for large buildings and organizations, enabling centralized monitoring and control of devices such as lights and fans. Built a web-based management platform using **Next.js, Express.js, and React Native**, integrated with **ESP32 and Arduino UNO microcontrollers programmed in C++** for hardware-level device control.

- Developed a centralized admin dashboard for managing devices, rooms, users, and access permissions across the building.
- Implemented room management and user access control to ensure authorized individuals could manage designated spaces and devices.
- Integrated IoT hardware with the software platform to enable remote control of connected electrical appliances.
- Designed a scalable management solution to support multiple rooms, users, and connected devices within large building environments.

Technologies: Next.js, Express.js, React Native, Arduino UNO, ESP32, C++, IoT, JavaScript.

COCO-VANTAGE

Next.js, NestJS, Prisma, PostgreSQL, Docker, GitHub Actions, Neon, Tailwind CSS

Enterprise-grade freelance web platform featuring high-throughput backend services, automated containerized deployments, and responsive analytics dashboards.

Kivi

Flutter, Dart, Docker, Github, NestJS, Node, PostgreSQL

Cloud POS-Subscription-Based Point-of-Sale System | Full-Stack Development

Designed and developed a cloud-based Point-of-Sale (POS) system for small and medium-sized enterprises (SMEs), offering a subscription-based business model. Built to support sales and inventory operations across devices, the platform enables businesses to manage products, track inventory, and add items using mobile phones or PCs, with barcode scanning and both online and offline functionality.

- Developed a cross-platform application using **Flutter** and backend services using **NestJS**, with **PostgreSQL** for data storage and management.
- Implemented inventory management features to manage products, stock levels, and item information.
- Designed role-based access management through an administrative panel, allowing business owners to grant appropriate access to multiple users.
- Supported product entry and barcode scanning using mobile phones, alongside PC-based workflows.
- Designed the system to support online and offline usage, enabling businesses to continue essential POS operations with limited or no internet connectivity.
- Integrated **GitHub Actions** into the development workflow to automate CI/CD tasks and streamline software delivery.

Technologies: Flutter, NestJS, PostgreSQL, GitHub Actions, REST APIs, Barcode Scanning, Cloud POS, Inventory Management, Subscription-Based SaaS.

Personal Projects

Smart Home IoT Management System

C++, Next.Js, ExpressJS, Node, PostgreSQL, React Native

Smart Home IoT Management System | Full-Stack & IoT Development

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SKILLS

Cloud & DevOps: AWS, Docker, Kubernetes, Ansible, GitHub Actions, GitLab, Git, Nginx, Apache
Frontend & Mobile: Next.js, React Native, Flutter, JavaScript, Dart, Tailwind CSS
Backend & Services: .NET, NestJS, Node.js, C#, PHP, C++, REST APIs
Databases & ORM: PostgreSQL, MySQL, MongoDB, Microsoft SQL Server, Prisma, Firebase
Tools & Scripting: PowerShell, Postman, Vercel, Canva, Linux

REFERENCES

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